Tidd: Augmented Tabletop Interaction Supports Children with Autism to Train Daily Living Skills

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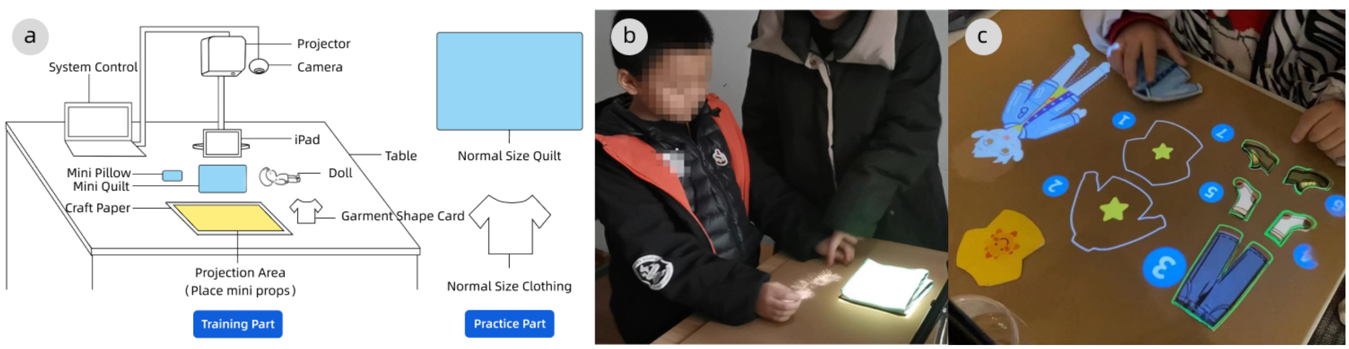

Figure 1: (a) System diagram of Tidd. (b) The training part of bed-making skills. (c)The training part of dressing skills.

ABSTRACT

Children with autism may have difficulties in learning daily living skills due to repetitive behavior, which poses a challenge to their independent living training. Previous studies have shown the potential of using interactive technology to help children with autism train daily living skills. In this poster, we present Tidd, an interactive device based on desktop augmented reality projection, designed to support children with autism in daily living skills training. The system combines storytelling with Applied Behavior Analysis (ABA) therapy to scaffold the training process. A pilot study was conducted on 13 children with autism in two autism rehabilitation centers. The results showed that Tidd helped children with autism learn bed-making and dressing skills while engaging in the training process.

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1 INTRODUCTION

Improving daily living skills (DLS) is one of the crucial objectives in interventions and training programs for children with Autism Spectrum Disorder (ASD). Teaching DLS to children with autism in their early developmental stages can promote independence and enhance their well-being [Autism Speaks [n.d.]]. Various interactive techniques have been proven effective in helping children with autism in DLS training, such as using sensors to help with shoe-wearing skill traning [Moorthy et al. 2016], and allow children with autism to learn shopping skill in virtual reality[Adjorlu et al. 2017]. Our study aims to investigate the potential and impact of using augmented reality projection technology in facilitating DLS training for children with autism. To achieve this goal, we developed Tidd, an interactive device based on augmented reality projection of tabletop interaction. We also designed two training tasks using Tidd to represent the training of bed-making skill and dressing skill. The contribution of this poster includes: 1) explored the potential of the augmented reality projection device Tidd in facilitating DLS training for children with autism, and 2) insights based on a pilot user study involving 13 children with autism.

2 SYSTEM DESIGN

2.1 Interactive Design

Our system hardware includes a projector, a camera, a tablet and some objects related to DLS training tasks (Figure 1a). The software was developed using the Unity platform, OpenCV and Halcon libraries are used to identify and match the objects. The interaction of Tidd is divided into two parts: the training part and the practice part. In the training part, targeted DLS skill is broken down into its simplest movement based on the Applied Behavior Analysis therapy, a common therapy in skill training for children with ASD. And system would use projected content to guide users to perform those movements. Specifically, users will interact with miniaturized objects (such as mini pillows, mini quilts, etc.) to learn corresponding DLS. In the practice part, participants perform operations on normal-size objects independently using the skill learned in the training part.

2.2 Training Task

According to the "Peizhi School Compulsory Education Curriculum Standards" [1] in China, we selected two DLS, bed-making and dressing. (A)Bed-making: In the training part, participants need to watch a demonstration animation of bed-making, and then practice the process twice using mini-beddings (miniature quilt and pillow) based on the projected content (Figure 1b). In the practice part, participants need to fold a normal-sized quilt. (B)Dressing: In the training part, system will project different cloth outlines of different seasons. Participants then need to choose and place the correct mini-clothing into the projection outlines to learn about commonly used clothes in different seasons. Then, they need to follow the number to move the mini-clothing to the projected character to learn the dressing sequence (Figure 1c). In the practice part, participants need to identify and pick out normal-sized clothing for a given season, and then place them in the correct dressing sequence.

3 USER STUDY

3.1 Procedure

This research was conducted under the supervision of the local University Institutional Review Board. We conducted a pilot user study at two developmental disability rehabilitation centers, where the experiment began after guardians signed a consent form. Thirteen children (ages 5-12) with varying degrees of autism volunteered to be participants in our study, and their guardians and therapists were invited to join the experiment as observers. Researchers provide assistance during the training part to help participants acclimate to our system. With permission from the rehabilitation center and consent from the guardians, we video recorded the entire experiment. After the experiment was completed, we interviewed guardians and therapists who had observed the entire experiment, focusing on the participants' performance and the differences between our approach and their traditional teaching methods. These interviews were recorded for analysis.

[1]http://www.moe.gov.cn/srcsite/A06/s3331/201612/W020180117596171784593.pdf

3.2 Results

Engagement: Based on the observation and video records from study, we found that participants would actively interact with and engage with the projected content in training part. 4 children attempted to read out the projected animation story when practice bed-making skill, 5 children actively touched the animations and cartoon patterns projected by the system. When complete the training part, four children exhibited positive behaviors such as laughter, clapping, and jumping. During the interview, both guardians and therapists highly appreciated this training method that combines visual stimulation and physical interaction, believing that it can improve children's engagement during training.

Effectiveness: All participants could complete training part in the first try, with five children could independently complete the practice part. After extra 1-2 sessions of the training part, all participants were able to independently complete the practice part. More specifically. In addition, 10 children were able to follow the guidance from our system and responded positively. Therapists pointed out in interviews that our system and training task was more interesting, instructive, and operational than the usual teaching method they use, which is hands-on guidance and verbal explanations.

3.3 limitation and future work

Our experiment was conducted during the Covid-19 pandemic, and due to health restrictions, the sample size was relatively small. In addition, the duration of the experiment made it difficult to track the long-term effects on children in their daily environments. In future work, we plan to continue following up with the participants, observe their learning process, and knowledge transfer from our interactive system to real-life practice. Furthermore, we are considering applying the our system to train more skills, such as brushing teeth and dining to evaluate the system's effectiveness on different daily living skills.

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